



Mariano Martín Fresno

Embedded Systems Engineer | IoT Developer | Machines & Systems | Electronics & Low-Level Programming

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Summary

Embedded Systems Engineer and IoT Developer with 15+ years of experience in electronics, low-level programming, machine control, and hardware-software integration. Strong foundation in C/C++, industrial systems, and embedded applications, combined with broader software development experience.

Experience

- Jan 2003 – Jan 2005 **Maintenance and Development Technician**, Machines & Systems Automation S.R.L. – Vicente López, Buenos Aires, Argentina
- Industrial automation, PLC programming, SCADA systems, and machine control for automotive and manufacturing industries.
- PLC programming and complete software development for automated filling stations and cutting systems
 - Design and execution of electronic devices for microcontroller programming and interfacing
 - SCADA systems: Siemens SINEC_H1 ring networks for remote monitoring of industrial processes
 - Human-machine interface (HMI) programming for automated equipment
 - Modernization and reconditioning of power and control panels for machine tools (CARRARO/TRACTORES DEUTZ)
 - Construction of automatic devices for manufacturing (ultrasonic welding, filling, cutting dies)
 - Installation and maintenance of industrial communication systems and telemetry
 - Electrical and pneumatic panel construction and cabling from engineering drawings
- Jan 2011 – Jan 2016 **CEO & Co-Founder**, Catálisis Desarrollos e Instalaciones – Colegiales, Capital Federal, Argentina
- Software and hardware development, IT infrastructure, and embedded systems for industrial applications.
- Embedded systems development with Arduino, C++, and Python
 - Hardware and software development for industrial applications
 - IT network installation and maintenance (servers, databases, VPN, PBX systems)
 - Development with Laravel, .NET, PHP, Qt, and Linux systems
 - Equipment provisioning and software implementations

- Jan 2009 – Feb 2010 **Technician in Teaching Laboratories**, Facultad de Ciencias Exactas y Naturales, UBA – Departamento de Física, FCEN, Buenos Aires, Argentina
Laboratory equipment maintenance, data acquisition systems, and instrumentation.
- Repair and modification of laboratory equipment (lasers, discharge lamps, transducers)
 - Data acquisition systems repair and LabView interface programming
 - Optical, acoustic, and electronic instruments maintenance
- Jan 2015 – Nov 2018 **Full Stack Engineer / Embedded Systems Developer**, d3 – Argentina
Development of embedded systems, machine control, and multimedia systems with focus on hardware integration.
- C++ multimedia systems development (audio streaming, DirectShow, ffmpeg)
 - Machine control systems and device integration
 - Low-level system programming and hardware integration
 - Real-time audio/video processing systems
 - Embedded system architecture and optimization
 - Cross-platform development for embedded devices and industrial systems
- Dec 2018 – July 2023 **Senior Software Engineer**, Avature – Argentina
Backend systems development with focus on performance, system-level programming, and machine-to-machine integrations.
- Apache NiFi custom processors in Java (data pipeline processing for machine data)
 - Integration framework development (system-level integrations and device connectivity)
 - Performance optimization and low-latency systems
 - Linux system programming and automation
 - System integration for distributed machine networks
- Aug 2025 – present **Senior Full Stack Developer**, Healthee – Buenos Aires, Argentina
Full-stack development with focus on system integration, diagnostics, and performance.
- System diagnostics and troubleshooting for integrated environments
 - Secure microservices for device connectivity and telemetry
 - Optimization of real-time data acquisition for healthcare applications
 - System integration with Docker and CI/CD for Linux-based targets
 - Workflow improvements for hardware-adjacent development and support

Education

- 1999 – 2005 **República Francesa**, Electronic Technician in Electronics and Communications – Argentina
Technical education in electronics, digital systems, communications, and embedded systems fundamentals.
- 2008 – 2012 **Universidad de Buenos Aires**, Licenciatura (40% completed) in Physical Sciences – Buenos Aires, Argentina
- 2012 – 2013 **Universidad de Buenos Aires**, Licenciatura (Introductory studies) in Philosophy – Buenos Aires, Argentina

Skills

Embedded & Low-Level Programming: C, C++, System Programming, Real-time Systems, Hardware Integration

Machines & Systems: PLC Programming, SCADA Systems, Industrial Automation, Machine Control, HMI Programming, Telemetry

Electronics & Hardware: Digital Electronics, Embedded Systems, Microcontrollers, Arduino, Signal Processing, Circuit Design, Hardware Repair

Multimedia & Streaming: FFmpeg, DirectShow, Audio/Video Processing, Codec Development

IoT & Connectivity: Embedded Linux, Device Integration, Protocol Implementation, M2M Communication

Programming Languages: C, C++, Python, Java, TypeScript, JavaScript

AI & Automation: Cursor, Antigravity, OpenCode, Codex, Ollama, GitHub Copilot, Prompt Engineering

System Tools: Linux, Git, Docker, CI/CD, Performance Optimization

Languages

Spanish (Native)

English (Professional Working)